Final Project Proposal

Reinforcement Learning-Based Observer Trajectory Design for Spacecraft Proximity Operations

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1 Problem Definition

1.1 Motivation and Background

Spacecraft proximity operations are central to emerging space missions such as On-Orbit Servicing (OOS), In-Space Assembly and Manufacturing (ISAM), debris inspection, and cooperative rendezvous. In most existing studies, the primary focus lies on guidance and control of a *chaser spacecraft* performing rendezvous or docking with a target.

However, in many practical missions, a *third observer spacecraft* is required to monitor such proximity maneuvers. Examples include:

- Safety supervision during autonomous docking
- Optical verification of servicing operations

Unlike the chaser, whose objective is typically to minimize docking error, the observer must maintain favorable viewing geometry of *two moving spacecraft* while respecting fuel limitations and safety constraints.

Designing such an observer trajectory is nontrivial because:

- The target and chaser follow time-varying relative trajectories.
- Visibility constraints are nonlinear and geometry-dependent.
- Fuel consumption must be minimized.
- Closed-form optimal solutions are generally unavailable.

This motivates a learning-based decision framework capable of handling multi-objective trade-offs and time-varying geometric constraints.

1.2 System Description

We consider three spacecraft in Low Earth Orbit (LEO):

- Target spacecraft (nominal circular orbit)
- Chaser spacecraft (executing predefined proximity maneuver)
- Observer spacecraft (decision-making agent)

All relative states are expressed in the LVLH frame centered at the target.

The chaser trajectory is assumed known (e.g., preplanned rendezvous profile), while the observer must adaptively position itself to maintain continuous visibility of both spacecraft.

1.3 Geometric Visibility Requirement

Let $\boldsymbol{r}^{O/T}$ and $\boldsymbol{r}^{C/T}$ denote observer and chaser positions relative to the target.

The observer has a conical field-of-view (FOV) characterized by:

- Half-angle θ_{FOV}
- Maximum sensing range $R_{\max}$

At time t, visibility is satisfied if both target and chaser lie within the observer FOV cone:

$$\angle\Big(\boldsymbol{r}^{T/O}, \boldsymbol{b}\Big) \leq \theta_{\text{FOV}}, \quad \angle\Big(\boldsymbol{r}^{C/O}, \boldsymbol{b}\Big) \leq \theta_{\text{FOV}},$$

and

$$\|\boldsymbol{r}^{T/O}\|, \|\boldsymbol{r}^{C/O}\| \leq R_{\max},$$

where $\boldsymbol{b}$ denotes the observer boresight direction.

Maintaining this dual-object visibility constraint over a finite horizon constitutes the primary mission objective.

1.4 Formal Problem Statement

Given:

- Predefined target and chaser trajectories
- HCW relative dynamics for the observer
- Field-of-view and range constraints
- Maximum allowable impulsive Δv

determine a sequence of impulsive maneuvers $\{a_0, a_1, \ldots, a_T\}$ that maximizes cumulative mission performance:

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{T} \gamma^t \left(w_1 \mathbf{1}_{\text{visibility}} - w_2 \|\boldsymbol{r}_t^{O/T}\| - w_3 \Delta v_t\right)\right]$$

subject to:

$$\text{HCW dynamics}$$
$$\Delta v_t \leq \Delta v_{\max}$$

1.5 MDP Formulation

We model this as a finite-horizon Markov Decision Process:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, T)$$

where:

- $\mathcal{S}$ includes observer and chaser relative states
- $\mathcal{A}$ is the set of feasible lattice waypoint transitions
- P is induced by discrete-time HCW dynamics
- R encodes visibility and fuel trade-offs
- T is the finite mission horizon

The decision problem is therefore a constrained, multi-objective, visibility-aware guidance design task under linearized orbital dynamics.

1.6 State Space $\mathcal{S}$

The state at time t is defined as:

$$s_t = \left(\boldsymbol{r}_t^{O/T}, \boldsymbol{r}_t^{C/T}, t\right)$$

where:

- $\boldsymbol{r}_t^{O/T}$: observer position relative to target
- $\boldsymbol{r}_t^{C/T}$: chaser position relative to target
- t : discrete time index

The observer position is discretized using a 3D lattice in the LVLH frame.

1.7 Action Space $\mathcal{A}$

At each time step, the observer selects a neighboring lattice waypoint:

$$a_t \in \mathcal{A}(s_t)$$

This implicitly determines an impulsive Δv maneuver to transition to the selected waypoint.

1.8 Transition Model

Orbital dynamics are approximated using Hill–Clohessy–Wiltshire (HCW) equations:

$$s_{t+1} = f(s_t, a_t) + \epsilon_t$$

where ϵ_t captures discretization and modeling uncertainty.

1.9 Hill–Clohessy–Wiltshire (HCW) Dynamics

Let the target spacecraft move on a circular orbit with mean motion n. In the LVLH frame centered at the target, the relative motion of the observer spacecraft $\boldsymbol{r} = [x\ y\ z]^\top$ satisfies the linearized Hill–Clohessy–Wiltshire equations:

$$\ddot{x} - 2n\dot{y} - 3n^2 x = u_x, \tag{1}$$

$$\ddot{y} + 2n\dot{x} = u_y, \tag{2}$$

$$\ddot{z} + n^2 z = u_z, \tag{3}$$

where:

- $n = \sqrt{\mu/a^3}$ is the mean motion of the target orbit,
- (x, y, z) are relative positions in the LVLH frame,
- (u_x, u_y, u_z) are control accelerations generated by impulsive Δv.

1.10 Reward Function

The reward is multi-objective:

$$R(s_t, a_t) = w_1 R_{\text{FOV}} - w_2 d_t - w_3 \Delta v_t$$

where:

- R_{FOV} : positive reward if both target and chaser lie within observer's field-of-view cone
- d_t : separation distance penalty
- Δv_t : maneuver cost

The objective is:

$$\max_{\pi} \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t) \right]$$

This explicitly places the problem within reinforcement learning.

2 RL Methods and High-Level Settings

We will implement and compare the following RL methods:

- Tabular Q-learning
- SARSA
- Policy Gradient

2.1 Learning Settings

- Finite horizon episodes
- ϵ-greedy exploration for Q-learning
- Learning rate α decay schedule
- Discount factor $\gamma \in (0, 1)$

Performance metrics:

- Cumulative reward
- Convergence speed
- Total Δv consumption
- Visibility maintenance ratio

3 Theoretical Background and Connection to Course Material

This project connects directly to key theoretical results covered in class:

3.1 Bellman Optimality Equation

For optimal policy π^*:

$$Q^*(s, a) = \mathbb{E}\left[R(s, a) + \gamma \max_{a'} Q^*(s', a')\right]$$

Tabular Q-learning converges to Q^* under:

- Sufficient exploration
- Diminishing learning rate
- Finite state-action space

Since our state space is discretized, these convergence guarantees approximately apply.

3.2 Policy Gradient Theorem

For parameterized policy π_θ:

$$\nabla_\theta J(\theta) = \mathbb{E}_\pi\left[\nabla_\theta \log \pi_\theta(a|s) Q^\pi(s, a)\right]$$

This provides a theoretical basis for continuous-policy extensions.

3.3 Difference from Standard RL Benchmarks

Unlike classical RL benchmarks (e.g., control of cart-pole), this problem:

- Has structured orbital dynamics
- Contains geometric visibility constraints
- Incorporates multi-objective reward trade-offs

Thus it serves as a domain-specific RL application with physics-based structure.

4 Advantages, Drawbacks, and Possible Improvements

4.1 Advantages

- Captures visibility-aware guidance problem
- Does not require closed-form solution
- Naturally handles multi-objective trade-offs

4.2 Drawbacks

- Curse of dimensionality with finer lattice
- Reward tuning sensitivity
- Approximation errors from discretization

4.3 Possible Improvements

- Continuous control (PPO / DDPG)
- Multi-agent RL formulation
- Robust MDP under orbital uncertainty
- Partial observability (POMDP extension)

5 Expected Outcome

We expect to:

- Demonstrate convergence of RL-based observer trajectory design
- Quantify trade-off between fuel consumption and visibility performance
- Compare theoretical convergence behavior with simulation results

6 Prior Work, Existing Theoretical Results, and Distinguishing Features

6.1 Prior Work on RL for Spacecraft Proximity Operations

Reinforcement learning (RL) has been increasingly investigated as a tool to learn closed-loop guidance (or integrated guidance and control) policies for spacecraft proximity operations, with the goal of reducing manual engineering effort and enabling autonomy under complex mission constraints. A representative example is the *deep guidance* concept, where deep RL is used to learn a *guidance* policy while a conventional controller is retained for trajectory tracking, explicitly aiming to mitigate simulation-to-reality (sim-to-real) issues [2]. In [2], the learned policy outputs velocity commands which are tracked by a conventional controller; the authors demonstrate that the policy can be trained entirely in simulation and transferred to a laboratory spacecraft testbed with comparable performance, emphasizing that separating guidance from control can reduce overfitting to imperfect simulated dynamics [2].

More recent work has expanded RL formulations to incorporate uncertainty, perception-driven objectives, and complex dynamics. For example, [3] formulates an active mapping / shape reconstruction guidance problem as a POMDP and evaluates PPO-based deep RL policies under noisy navigation measurements. This work explicitly defines a visibility model based on camera field-of-view (FOV) and incidence-angle constraints on a triangular mesh target, and assesses robustness via extensive training/testing campaigns [3].

In cislunar environments, meta-RL has also been proposed to address sample inefficiency and generalization limitations of standard deep RL. In [4], an LSTM-based meta-RL agent trained with PPO is used for real-time optimal guidance (and control) in the CR3BP setting, while enforcing operational constraints such as an approach corridor modeled by a truncated cone (frustum) to ensure safe docking [4].

6.2 Existing Theoretical Results for the Proposed RL Algorithms

The proposed project focuses on tabular Q-learning, SARSA, and REINFORCE. These algorithms have well-established theoretical foundations under standard MDP assumptions:

(1) Q-learning (off-policy TD control). For a finite MDP, tabular Q-learning converges to the optimal action-value function $Q^\star$ with probability one when: (i) all state–action pairs are visited infinitely often (sufficient exploration, e.g., GLIE), (ii) the learning rate satisfies the Robbins–Monro conditions $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$, and (iii) rewards are bounded [1]. The main idea is that the Bellman optimality operator is a contraction in the max-norm, and the stochastic approximation updates track this contraction mapping despite sampling noise.

(2) SARSA (on-policy TD control). Tabular SARSA converges to Q^π for the behavior policy π under analogous diminishing step-size and sufficient exploration conditions; in the GLIE limit (policy becomes greedy while maintaining infinite exploration), SARSA converges to an optimal policy in the tabular setting [1]. The key idea is that SARSA performs policy evaluation and improvement *on-policy*, which can be more stable under certain exploration schemes and constraint-heavy environments.

(3) Monte-Carlo policy gradient For differentiable stochastic policies $\pi_\theta(a|s)$, policy gradient provides an *unbiased* estimator of the policy gradient $\nabla_\theta J(\theta)$ based on the likelihood-ratio identity

[1]. The main idea is that $\nabla_\theta \mathbb{E}[G] = \mathbb{E}[\nabla_\theta \log \pi_\theta(a|s)\, G]$, so gradient ascent is possible without differentiating the environment dynamics. While unbiased, the estimator can have high variance, motivating baselines/advantages (not required in this project, but a natural extension).

6.3 What Course Material Directly Helps This Project

This project directly leverages core course concepts: (i) formal MDP modeling (Markov property, finite-horizon episodic returns, discounting), (ii) dynamic programming and Bellman equations (fixed-point structure that motivates TD learning), (iii) contraction mappings and stochastic approximation intuition for why tabular TD methods converge, and (iv) policy gradient theorem / likelihood-ratio methods enabling gradient-based learning without known dynamics [1]. These principles justify the algorithm choices, define appropriate evaluation metrics (return, stability of learning curves), and guide reward shaping and exploration strategies.

6.4 Differences Between Our Simulation and Prior Simulation Setups

Although our work is inspired by proximity-operation RL literature, the *simulation objective, agent role, and constraints* differ in fundamental ways.

Difference vs. deep guidance docking simulations. The deep guidance framework in [2] is centered on a *chaser* performing pose tracking and docking, where the learned policy outputs velocity commands and a conventional controller tracks them; their proof-of-concept environment is designed to match a planar laboratory testbed and uses simplified planar dynamics for experimental validation [2]. In contrast, our simulation focuses on a *third observer spacecraft* as the decision-making agent, operating in the LVLH frame to maintain visibility of *two* spacecraft (target and chaser) under geometric FOV constraints, while explicitly penalizing fuel usage (Δv) and separation. Furthermore, we use an impulsive maneuver abstraction on a discretized 3D lattice, which yields a finite state–action MDP suited for tabular RL.

Difference vs. vision-driven fly-around / mapping simulations. The fly-around mapping work [3] addresses a chaser trajectory planning problem to maximize target shape reconstruction quality and introduces a detailed visibility model (FOV and incidence-angle constraints on a triangular mesh), with the decision problem framed as a POMDP under navigation uncertainty and solved using PPO [3]. Our simulation does *not* aim at reconstructing target geometry; instead, we impose a task-level requirement that *both* target and chaser must remain inside the observer FOV, which is a multi-object visibility maintenance objective rather than face/mesh coverage. Additionally, we prioritize a tabular baseline comparison (Q-learning/SARSA/Policy Gradient) rather than continuous-control PPO, to study learning behavior in a controlled finite MDP.

Difference vs. cislunar meta-RL simulations. The meta-RL study [4] considers cislunar relative motion in CR3BP and emphasizes robustness/generalization via LSTM-based meta-learning, while enforcing docking safety constraints such as a frustum approach corridor [4]. Our setting is LEO LVLH with HCW dynamics approximation and discrete impulsive waypoint selection. We also treat visibility maintenance as the primary mission constraint, rather than approach corridor docking constraints.

6.5 Key Distinguishing Features of the Proposed Project

The proposed project is distinguished by the following aspects:

- **Observer-centric decision making:** Unlike most RL proximity-operation studies that optimize the chaser's docking or fly-around behavior, our RL agent is a *third observer* spacecraft whose task is to maintain persistent visibility of a proximity-operation pair (target–chaser) under FOV and distance constraints.
- **Finite MDP via 3D LVLH lattice discretization:** We deliberately discretize the observer position into a 3D lattice, producing a finite state–action space where tabular Q-learning and SARSA theoretical guarantees are the most directly applicable [1].
- **Explicit multi-objective trade-off:** The reward balances (i) dual-object visibility satisfaction, (ii) separation penalties, and (iii) impulsive Δv cost, enabling quantitative analysis of the visibility–fuel Pareto trade space.
- **Didactic baseline study with controlled physics:** By using HCW dynamics with discretization noise, we create a physics-structured yet finite MDP suitable for a controlled comparison of Q-learning, SARSA, and REINFORCE learning behavior, before extending to continuous control or partial observability.

7 Discussion: Advantages, Drawbacks, and Theoretical Guarantees of the RL Methods

7.1 Advantages and Theoretical Guarantees

(1) Tabular Q-learning Advantages:

- Model-free: does not require explicit knowledge of transition probabilities.
- Off-policy: can learn optimal policy independent of behavior policy.
- Strong convergence guarantees in finite MDPs.

Theoretical guarantee: In a finite MDP, Q-learning converges to the optimal action-value function $Q^\star$ with probability one if: (i) all state-action pairs are visited infinitely often, (ii) the learning rate satisfies Robbins–Monro conditions, (iii) rewards are bounded.

This is supported by the contraction property of the Bellman optimality operator.

Limitation in this project: The convergence guarantee strictly holds only for finite state-action spaces. As lattice resolution increases, the state space grows exponentially, leading to slow convergence and practical sample inefficiency.

(2) SARSA Advantages:

- On-policy learning can provide more stable behavior under constraint-heavy settings.
- Naturally incorporates exploration strategy into policy evaluation.

Theoretical guarantee: Under diminishing step size and sufficient exploration (GLIE), tabular SARSA converges to an optimal policy in finite MDPs.

Limitation in this project: On-policy learning can converge more slowly than off-policy Q-learning. Additionally, performance depends strongly on exploration schedule.

(3) Policy Gradient Advantages:

- Direct optimization of expected return.
- Naturally extendable to continuous action spaces.
- Does not require value-function estimation.

Theoretical guarantee: Policy Gradient provides an unbiased estimator of the policy gradient, ensuring convergence to a local optimum under suitable step-size conditions.

Limitation in this project: The estimator has high variance, which may result in unstable learning. Convergence is only guaranteed to a local optimum, and performance depends on reward scaling and variance reduction.

7.2 General Drawbacks in the Context of This Problem

- **Curse of dimensionality:** Finer LVLH lattice discretization significantly increases the state space.
- **Reward sensitivity:** Multi-objective weighting (w_1, w_2, w_3) strongly affects learned policy.
- **Sample inefficiency:** Orbital dynamics evolve over long horizons, requiring many episodes.
- **No explicit constraint guarantees:** Standard RL optimizes expected reward but does not strictly enforce safety or visibility constraints.

7.3 Suggestions for Improvement

(1) Baseline / Advantage Methods For policy gradient methods, introducing a state-value baseline or advantage estimation can significantly reduce gradient variance and improve convergence stability.

(2) Constrained MDP Formulation Instead of encoding visibility as soft reward, the problem can be reformulated as a Constrained MDP (CMDP), where visibility violation probability is bounded. This provides stronger theoretical interpretation of safety guarantees.

(3) Actor-Critic Methods Combining value-function approximation with policy optimization (e.g., actor-critic) can improve sample efficiency compared to pure Policy Gradient.

(4) Continuous Control Extension Replacing lattice-based impulsive control with continuous thrust control and applying PPO or DDPG would improve scalability and remove discretization-induced suboptimality.

(5) Model-Based Acceleration Since orbital dynamics are known (HCW), a model-based RL or hybrid dynamic programming approach could reduce sample complexity and improve convergence speed.

References

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